Standard Access Control List (ACL)

Objective

The objective of this lab is to implement a **Standard Access Control List (ACL)** to restrict network access for a **specific host** while allowing other devices to communicate normally across networks.

This demonstrates how administrators can enforce **basic security policies** by controlling which devices are allowed or denied access based on their **source IP address**.

Network Topology

The network consists of:

- **One Cisco 2911 router** connecting two separate LANs
- **Two switches (Cisco 2960)** for local connectivity
- **Four PCs**, two in each network

Network Layout:

- **LAN 1 (Left Side)** connected to Router interface
- **LAN 2 (Right Side)** connected to another router interface
- The router acts as the **gateway and traffic filter (ACL device)**

IP Addressing Scheme

LAN 1 (Source Network)

- Network: 192.168.1.0/24

Device	IP Address
PC1	192.168.1.10
PC2	192.168.1.20

LAN 2 (Destination Network)

- Network: 192.168.2.0/24

Device	IP Address

PC3	192.168.2.10
PC4	192.168.2.20

Project Description

In this lab, a **Standard ACL** is implemented on the router to control traffic flowing from **LAN 1 to LAN 2**.

The ACL is designed to:

- **Deny one specific PC** from accessing the destination network
- **Allow all other PCs** to communicate normally

This shows how a simple ACL can enforce **host-level access control**.

ACL Policy

- **PC1 (192.168.1.10)** → ❌ Denied access to LAN 2
- **PC2 (192.168.1.20)** → ✅ Allowed
- **All devices in LAN 2** → Accessible to permitted hosts

ACL Placement

The Standard ACL is placed:

➡️ **On the router interface closest to the destination network (LAN 2)**

Reason:

- Prevents unnecessary blocking of traffic to other networks
- Ensures only traffic destined for LAN 2 is filtered

How the ACL Works

1. A PC from LAN 1 sends traffic toward LAN 2
2. The router receives the packet
3. The ACL checks the **source IP address**
4. If the source is **192.168.1.10 (PC1)**:
    - The packet is **dropped**

5. If the source is any other IP:
    - The packet is **forwarded normally**

Expected Behavior

Source Device	Destination	Result
PC1 (192.168.1.10)	LAN 2	❌ Blocked
PC2 (192.168.1.20)	LAN 2	✅ Allowed
PC3 / PC4	Internal LAN	✅ Normal communication

Verification

To confirm correct ACL operation:

- Attempt to **ping from PC1 to PC3 or PC4**
    - Result: ❌ Request fails
- Attempt to **ping from PC2 to PC3 or PC4**
    - Result: ✅ Successful

This confirms that:

- The ACL is correctly denying only the specified host
- Other traffic is not affected

Protocols and Concepts Used

- **Standard ACL** – Filters traffic based on source IP
- **IPv4** – Logical addressing
- **ICMP** – Used for testing connectivity
- **Routing** – Enables communication between networks

Conclusion

This lab demonstrates how a **Standard ACL** can be used to restrict access for a specific host while allowing other users to communicate freely. It highlights the simplicity and effectiveness of ACLs in implementing **basic network security policies**.